Sudhansu Sekhar Sahoo

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Summary

PL/SQL Developer with 3 years of experience in database development and production support, specializing in troubleshooting, incident resolution, and performance optimization. Experienced in handling system issues, working with stakeholders. Strong problem-solving skills with the ability to diagnose and resolve technical issues efficiently.

Education

Bachelor of Technology – Computer Science 08/2019 – 04/2023
GIET University CGPA: 8.29

Skills

Programming: Python, SQL, PL/SQL, Java (Basic)
Backend Development: FastAPI, REST APIs, Async Programming
Databases: Oracle 12c/19c, MySQL, PostgreSQL, SQLite
AI & LLM: Prompt Engineering, Retrieval-Augmented Generation (RAG), LLMs, Document Parsing, Text Chunking, Embeddings, Semantic Search
Database & Performance: EQuery Optimization, Execution Plan Analysis, Indexing, Stored Procedures, Functions, Packages, Triggers
Vector Databases: ChromaDB (Learning), Vector Search
Performance: Query Optimization, Execution Plan Analysis, Indexing,Debugging
Tools: Git, GitHub, GitLab CI/CD, Jira, SVN, XML, VS Code **Concepts:** ETL, Data Modeling, SDLC, Agile/Scrum, Production Support, Unix/Linux, Shell Scripting, Debugging

Experience

Tech Mahindra Sept 2023 – Present

- Developed and maintained PL/SQL procedures, functions, packages, triggers, and views to implement business logic.
- Optimized complex SQL queries and PL/SQL code, reducing execution time by up to 35% using indexing and execution plan analysis.
- Performed advanced query tuning, analyzed execution plans, and improved database performance and scalability.
- Handled incident management and prioritized issues based on business impact, ensuring timely resolution as per SLA.
- Monitored production systems, identified bottlenecks, and resolved performance-related issues.
- Worked on applications supporting telecom operations for Telefónica Germany (O2).
- Supported database operations and production activities in a telecom domain environment
- Worked with enterprise-scale telecom data and backend systems
- Participated in UAT and supported production deployments and release activities.
- Worked in Unix/Linux environments for deployment support, troubleshooting, and execution of database-related activities.
- Participated in full SDLC, including requirement analysis, development, testing, deployment, and support.
- Maintained technical and process documentation for database changes and enhancements.
- Used shell commands and automation scripts to support production processes and monitoring activities.
- Actively participated in Agile ceremonies including sprint planning, stand-ups, and retrospectives.

Projects

AI-Powered SQL Generator *Python, Streamlit, SQLite, Google Gemini API*

- Developed an AI-powered application that converts natural language prompts into optimized SQL queries using Google Gemini API.
- Engineered effective prompts to generate accurate, context-aware SQL statements for different database operations.
- Built an interactive web interface using Streamlit for seamless query generation and execution.
- Integrated SQLite as the backend database for query execution, validation, and testing.
- Implemented SQL validation mechanisms to prevent unsafe or malformed queries before execution.
- Automated SQL generation workflows, significantly reducing manual query-writing effort and improving developer productivity.
- Applied database optimization concepts to generate efficient SQL queries with improved readability and performance.
- Utilized Git for version control and maintained project documentation to support collaborative development.

- Improved overall query generation efficiency by approximately 50% while ensuring reliable and secure database interactions.

Enterprise Knowledge Assistant (Production-Grade RAG System) Personal Project

- Designed and developed a production-ready Retrieval-Augmented Generation (RAG) application using Python and FastAPI for enterprise document search.
- Built REST APIs for document upload, validation, parsing, chunking, embedding generation, vector indexing, and AI-powered question answering.
- Implemented secure document ingestion supporting PDF, DOCX, TXT, and Markdown formats with automatic validation and structured storage.
- Developed a document processing pipeline to extract text, generate semantic chunks, and prepare documents for efficient retrieval.
- Generated vector embeddings and stored them in ChromaDB to enable high-performance semantic search across enterprise knowledge bases.
- Integrated Ollama-hosted Large Language Models with LangChain to answer user questions using retrieved document context.
- Designed Retrieval-Augmented Generation (RAG) workflow combining vector similarity search with LLM inference to improve response accuracy and reduce hallucinations.
- Implemented structured logging, centralized configuration, modular service architecture, and exception handling following production-grade backend development practices.
- Optimized retrieval quality using configurable chunk sizes, overlap strategies, and embedding pipelines to improve search relevance.
- Followed clean architecture principles by separating API routes, business logic, parsing services, chunking services, vector database services, and AI services.
- Version controlled the complete project using Git and GitHub while following Agile development practices and iterative feature delivery.

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations
- Oracle Certified MySQL Administrator
- Microsoft Azure Fundamentals (AZ-900)
- Cisco AI Technical Practitioner certification
- Standing Ovation Award-Tech Mahindra